

PHASE 6 – PROJECT TESTING PHASE

Project Title: FitBuddy – AI Fitness Plan Generator using Gemini Models

Team Members: Vennila, Varsha, Vijayalaksmi, Vaishnavi

1. Testing Overview

The testing phase verifies that FitBuddy accepts required inputs, processes requests correctly, displays generated plans clearly and handles normal errors appropriately.

2. Functional Test Cases

Test Case	Expected Result
Enter all required inputs and generate a plan	Application accepts the data and requests a fitness plan.
Leave a required input empty	Application shows a validation message and does not proceed normally.
Select different fitness goals	The generated request reflects the selected goal.
Change activity level	The generated request reflects the selected activity level.
Change workout days	The generated plan request reflects the selected number of days.
Provide optional preferences	Additional preferences are included when supplied.
Gemini/API request succeeds	Generated response is displayed in the result section.
Network/API request fails	A clear error message is shown instead of a broken interface.
View result on a smaller screen	Main interface remains readable and usable where responsive support is implemented.

3. Testing Types

- Functional testing – verify each main feature.
- Input validation testing – verify required-field handling.
- UI/usability testing – verify readability and ease of use.
- Integration testing – verify application-to-Gemini communication.
- Error-handling testing – verify normal API/network failure handling.
- Compatibility testing – check supported browsers/devices where applicable.

4. Test Result Recording

For each test case, the team should record the actual result, date, tester and status (Pass/Fail). Any failed case should be corrected and tested again.

5. Security Checks

- Confirm that API credentials are not committed to the public GitHub repository.
- Use configuration/environment variables for private credentials where supported.
- Do not include personal user information in public test screenshots or documentation.

6. Testing Conclusion

Testing should provide evidence that the implemented FitBuddy workflow behaves as intended. Final Pass/Fail results should be filled in from the team's actual test execution rather than assumed in advance.